Error in JW mapping with z2symmetry_reduction when running VQE-UCCS

https://github.com/qiskit-community/qiskit-nature/issues/241

ROW 35

Error in JW mapping with z2symmetry_reduction when running VQE-UCCSD #241

New issue

Closed

tanvipg opened on Jun 17, 2021

Information

- **Qiskit Nature version:** qiskit-nature: 0.1.3
- **Python version:** Python 3.8.10
- **Operating system:** macOS BigSur

What is the current behavior?

When implementing VQE-UCCSD with JW Mapping with z2symmetry_reduction; HF state returns None object. Similar behavior observed for Parity Mapping with z2symmetry_reduction and no two qubit reduction.

Steps to reproduce the problem

```
from qiskit import *
from qiskit_nature.drivers import PySCFDriver,UnitsType,HFMethodType,QMolecule
from qiskit_nature.problems.second_quantization import ElectronicStructureProblem
from qiskit_nature.converters.second_quantization import QubitConverter
from qiskit_nature.mappers.second_quantization import JordanWignerMapper,ParityMapper
from qiskit_nature.transformers import FreezeCoreTransformer
from qiskit_nature.circuit.library import HartreeFock,UCCSD
from qiskit_nature.algorithms import VQEUCCFactory,GroundStateEigensolver
from qiskit.algorithms.optimizers import L_BFGS_B,COBYLA
from qiskit.algorithms import MinimumEigensolver,VQE
from qiskit.utils import QuantumInstance
from qiskit.circuit.library import TwoLocal
from qiskit.opflow import PauliSumOp,TaperedPauliSumOp,Z2Symmetries,PauliOp

driver = PySCFDriver(atom="Li 0 0 0; H 0 0 1.4",
unit=UnitsType.ANGSTROM,
charge=0,
spin=0,
basis='sto-3g',
hf_method=HFMethodType.RHF)

es_problem = ElectronicStructureProblem(driver,[FreezeCoreTransformer()])
second_q_op = es_problem.second_q_ops()
num_particles = es_problem.num_particles
num_spin_orbitals = 2*es_problem.molecule_data_transformed.num_molecular_orbitals
print('Num particles:',num_particles)
print('Num molecular orbitals:',int(num_spin_orbitals/2))

qubit_conv_JWT = QubitConverter(mapper=JordanWignerMapper(),two_qubit_reduction = False, z2symmetry_reduction =
'auto')
qubit_op_JWT = qubit_conv_JWT.convert(es_problem.second_q_ops()[0], num_particles = num_particles, sector_locator =
es_problem.symmetry_sector_locator)

vqe_solver = VQEUCCFactory(quantum_instance=quantum_instance,
optimizer= opt,
initial_point= None,
gradient = None,
expectation = None,
include_custom = False,
ansatz = None, #default is UCCSD
initial_state = None)
```

Assignees

No one assigned

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

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Participants

include_custom = False,
ansatz = None, #default is UCCSD
initial_state = None)

calc = GroundStateEigensolver(qubit_conv_JWT, vqe_solver)
res = calc.solve(es_problem)
print(res)

What is the expected behavior?

Tapered and transformed HF state should be returned, which is not None object

Suggested solutions

Rukhsan on Jun 26, 2021

Please assign this issue to me.

Brogis1 on Aug 26, 2021 Contributor

I think it relates to issue #342 ... I have the problem only for homonuclear molecules though.

mrossinek on Nov 9, 2021 Member

@Brogis1 was right. This does indeed appear to be fixed, too, now that we closed #342.

After fixing the above snippet by specifying a `quantum_instance` and `opt` (which were missing) I was successfully able to run the simulation without running into errors. Below is the output using `opt = None` (which falls back to `SLSQP` in Terra) and `quantum_instance = QuantumInstance(Aer.get_backend("aer_simulator_statevector"))` on the latest Terra+Nature:

```
=== GROUND STATE ENERGY ===

* Electronic ground state energy (Hartree): -9.012181741489
  - computed part:      -1.121390258355
  - FreezeCoreTransformer extracted energy part: -7.890791483133
~ Nuclear repulsion energy (Hartree): 1.133951166257
> Total ground state energy (Hartree): -7.878230575231

=== MEASURED OBSERVABLES ===

  0:  # Particles: 2.000 S: 0.000 S^2: 0.000 M: 0.000

=== DIPOLE MOMENTS ===

~ Nuclear dipole moment (a.u.): [0.0  0.0  2.64561657]

  0:
  * Electronic dipole moment (a.u.): [0.0  0.0  4.47392393]
    - computed part:      [0.0  0.0  4.47710745]
    - FreezeCoreTransformer extracted energy part: [0.0  0.0  -0.00318352]
  > Dipole moment (a.u.): [0.0  0.0  -1.82830736]  Total: 1.82830736
                 (debye): [0.0  0.0  -4.64709334]  Total: 4.64709334
```

The qubit operator obtained for this problem with the JW mapping and automatic Z2 symmetry reduction required 6 qubits. The result above is numerically identical to that obtained with the full 10-qubit operator when no symmetry reductions are used.

Therefore, I am closing this issue 👍